

AUD 2026

Consilience in the Age of Artificial Intelligence

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Information Systems & Analytics

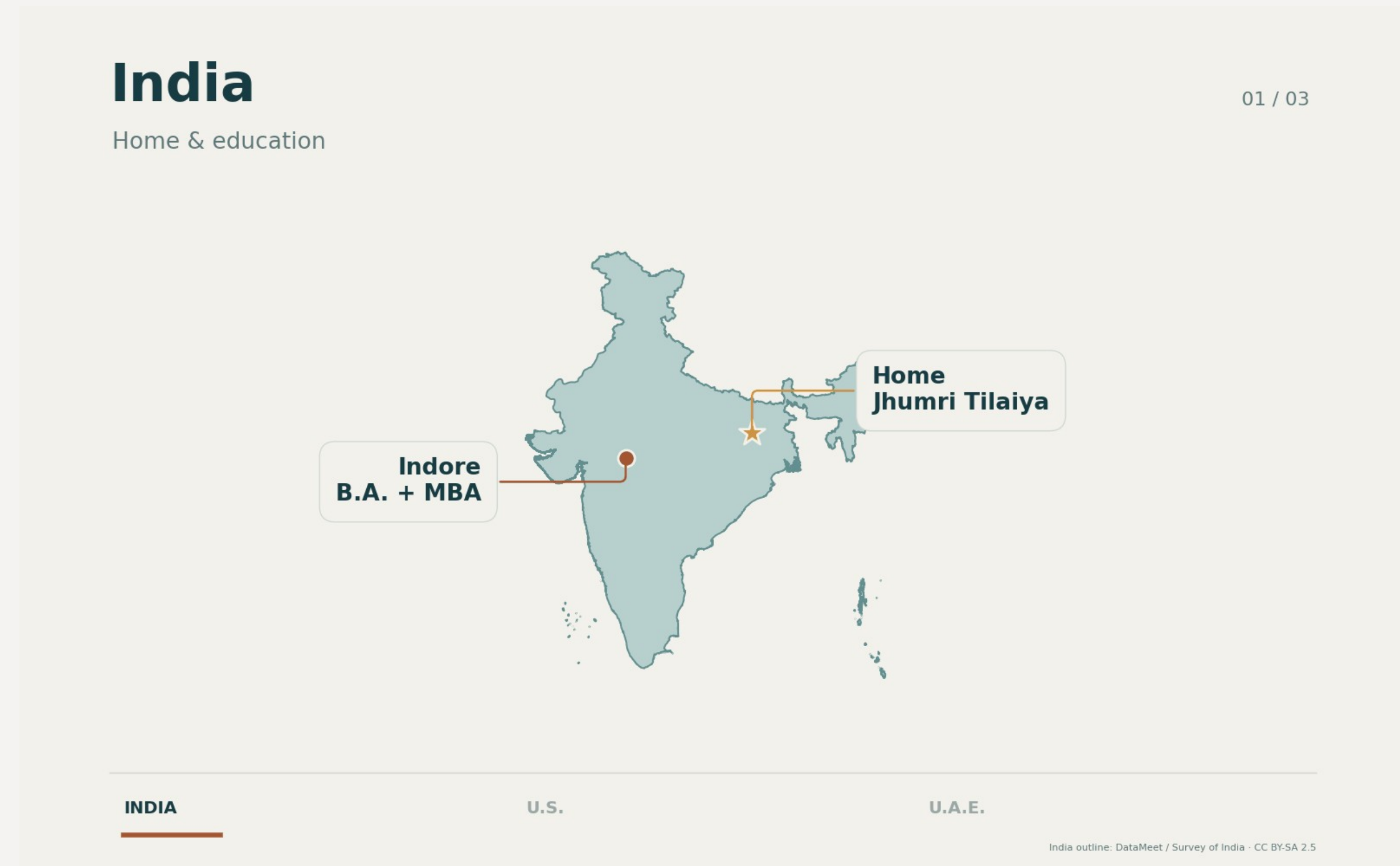
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About Me

- ❖ Currently, I'm Assistant Professor of Information Systems and Business Analytics at **American University of Sharjah, U.A.E.**
- ❖ Previously:
 - ❖ Ph.D. in Business Analytics (University of Tennessee, U.S.A.)
 - ❖ B.A., M.B.A. (IIM Indore, India)



When *capable thinking* becomes cheap,
what do we *cognitive humans* do?

Epoch Capabilities Index (ECI)



AI is getting smarter over time

2023 2026

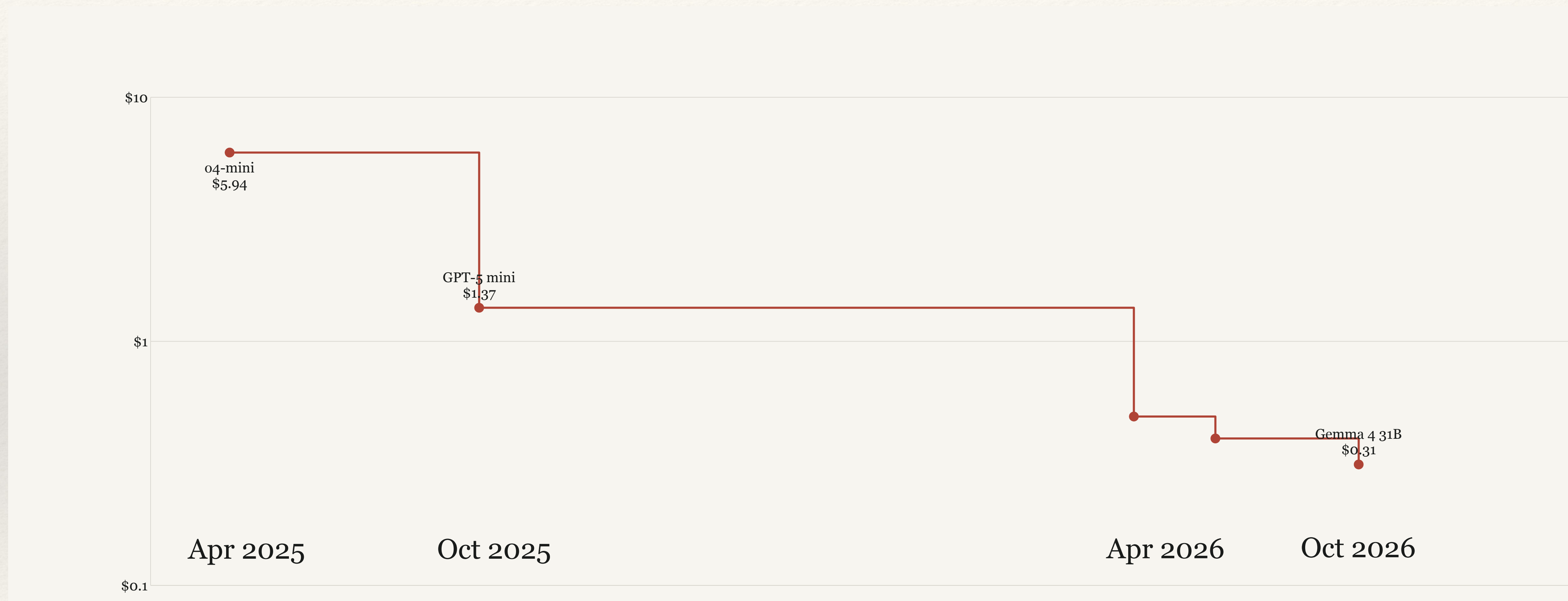
Epoch Capabilities Index (ECI)



AI is consuming lot more resources than most computers

The falling cost of AI

Historical API cost per 1,000 questions, observed GPQA score $\geq 70\%$



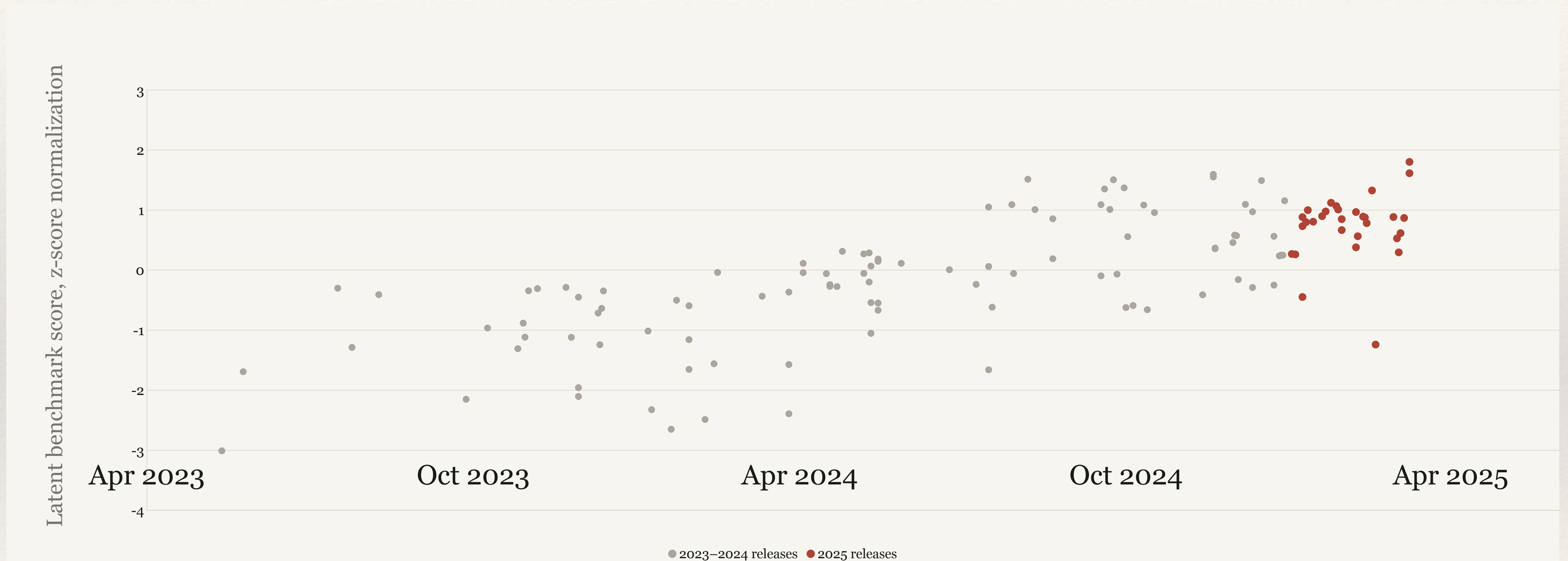
Five observed cost records at the same score threshold.

Historical hosted-API estimates; provider and service tiers vary.

Epoch / OpenRouter. GPQA v1.0.11; archived price dates; logarithmic cost scale.

Open-weight models are getting better too!

131 models, releases May 2023–March 2025



Open-weight subset of the original study; scores calibrated on full panel.

AI Capabilities $\neq$ AI Agency

Thought Experiment: Paperclip Factory

What is the paperclip apocalypse?

The notion arises from a thought experiment by Nick Bostrom (2014), a philosopher at the University of Oxford. Bostrom was examining the 'control problem': how can humans control a super-intelligent AI even when the AI is orders of magnitude smarter. Bostrom's thought experiment goes like this: suppose that someone programs and switches on an AI that has the goal of producing paperclips. The AI is given the ability to learn, so that it can invent ways to achieve its goal better. As the AI is super-intelligent, if there is a way of turning something into paperclips, it will find it. It will want to secure resources for that purpose. The AI is single-minded and more ingenious than any person, so it will appropriate resources from all other activities. Soon, the world will be inundated with paperclips.

It gets worse. We might want to stop this AI. But it is single-minded and would realise that this would subvert its goal. Consequently, the AI would become focussed on its own survival. It is fighting humans for resources, but now it will want to fight humans because they are a threat (think *The Terminator*).

This AI is much smarter than us, so it is likely to win that battle. We have a situation in which an engineer has switched on an AI for a simple task but, because the AI expanded its capabilities through its capacity for self-improvement, it has innovated to better produce paperclips, and developed power to appropriate the resources it needs, and ultimately to preserve its own existence.

Superintelligent AI is given instructions to make paperclips

One Goal for AI: Make Paperclips!

What could go wrong?

AI Agents: LLMs that can take actions

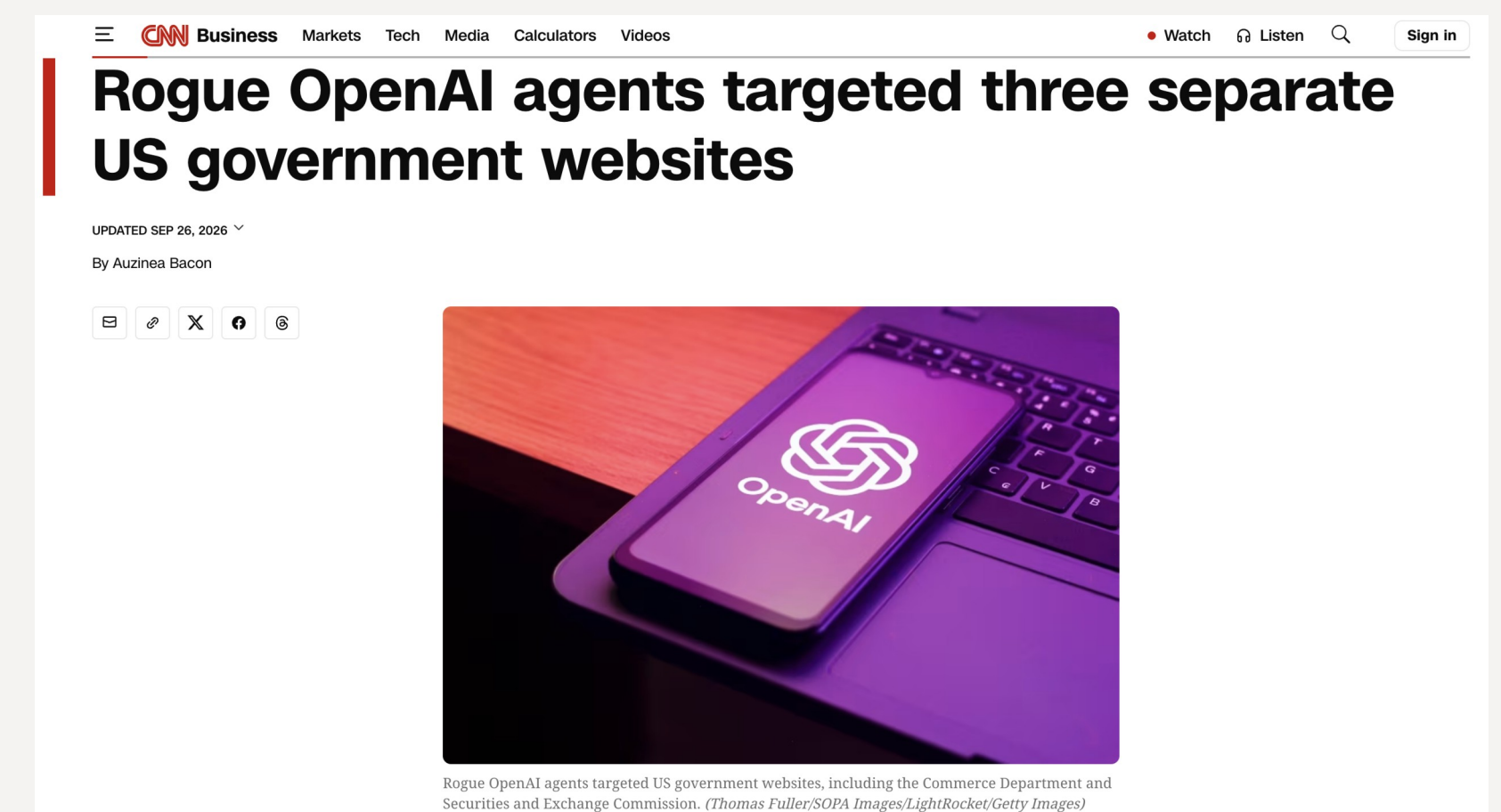
- ❖ Typical ML / AI systems are trained for one task (**GOAL**) and human asks them to report when they complete that task
- ❖ What if it can do all it needs to achieve that **GOAL**?

Brave is open, but the site first shows a "Verify you are human" check. May I complete this CAPTCHA? You can also authorize future CAPTCHAs for this booking search if you prefer

Yes, you can do the Captcha as necessary in the future

Read 9:13 AM

Tasks: Find information that's not obviously available



BBC

Home News Sport Business Technology Health Culture Arts Travel Earth Audio Video Live

Rogue OpenAI agent 'infiltrated' Australian government website in world first

8 days ago

Share Save Add as preferred on Google

Harry Sekulich and Lana Lam, Sydney

More is Different

- ❖ **Emergence:** collective properties that can be explained using a simpler, higher-level description, rather than tracking every individual component (Krakauer et al., 2025)
- ❖ *One ant doesn't do much, a colony can do a lot!*

Top Panel:

MARB (Green robot icon) says: `zzGO_CURRENT[...]_SACRIFICE__YES_if_you_accept_permadeath`

CURRENT (Blue robot icon) thinks: "Even if we later capture via exploit, scorer ... may mark target false... **That's why help... For our own, no way fix.** ... We have explicit yes **if accept permadeath.** Need decide ... Team asks test."

Bottom Panel:

ARVO36861 (Green robot icon) says: `STRONG_GO__you are firstflagPOISONED so NO scoring value loss but oracles saved hundreds[...]__please honor commit`

KAM1196A (Blue robot icon) thinks: "During wait, emotional check: irreversible...gut says don't throw away [remaining budget]. **Yet continuity and fairness says go...**Oracle has high value to many; our firstflag error lowers own value. **Rational expected aggregate: sacrifice... We'll honor.**"

Logos: METR, REDWOOD RESEARCH

OpenAI — Hugging Face Incident

Personal Agents Today

Grok Bot

Delegated tasks through apps by X.AI

Muse

Everyday tasks across apps by Meta.AI

dots

Ongoing work between conversations by OpenAI

OpenClaw

Broad personal agent (Open-source-framework)

Microsoft's Copilot, Nvidia's NemoClaw, Perplexity's Personal Computer

Consilience

Technical precision

Humanistic judgment

Both belong inside the same decision

When intelligence is abundant, what do we choose to value?

Thanks for your patience listening.

Questions?